

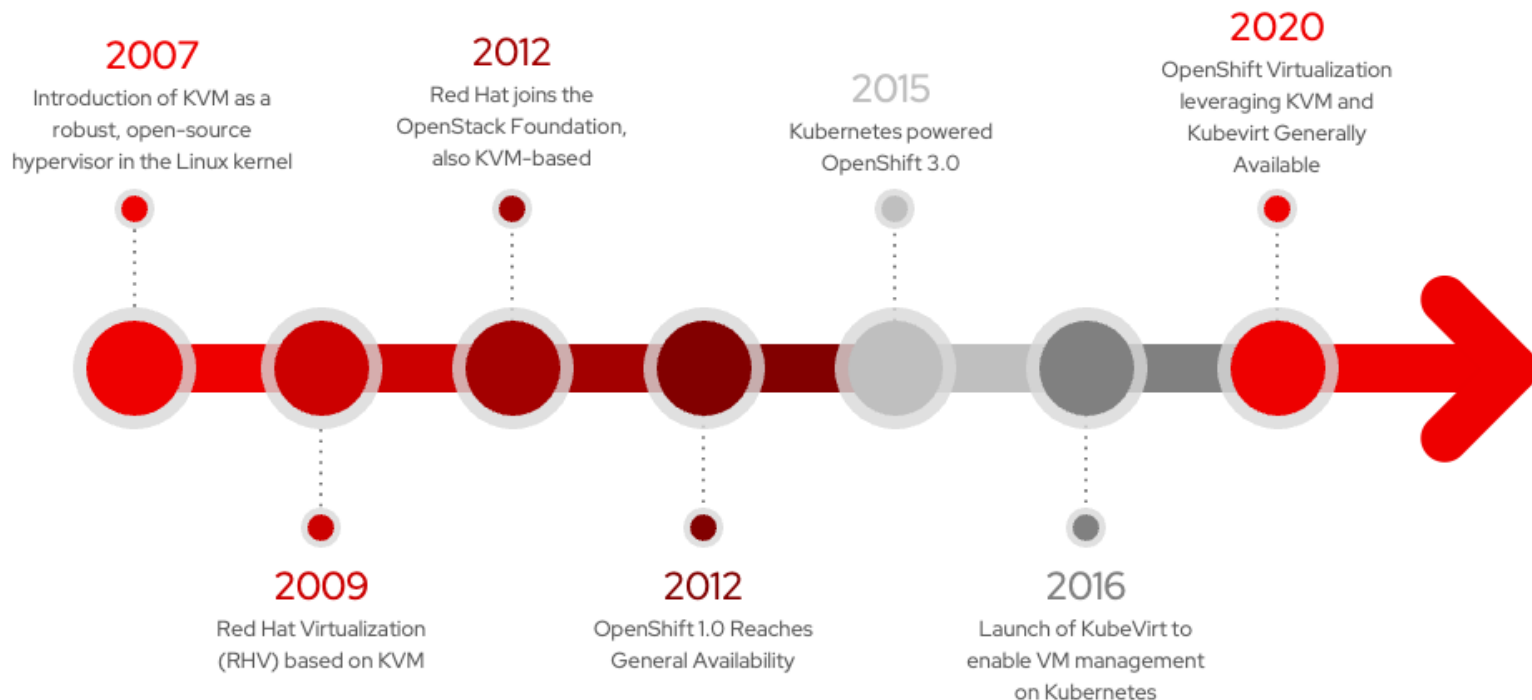


**RELAX,
WE CARE**



REDHAT OPENSIFT VIRTUALIZATION

Red Hat has a long history with Virtualization





VMs



Containers

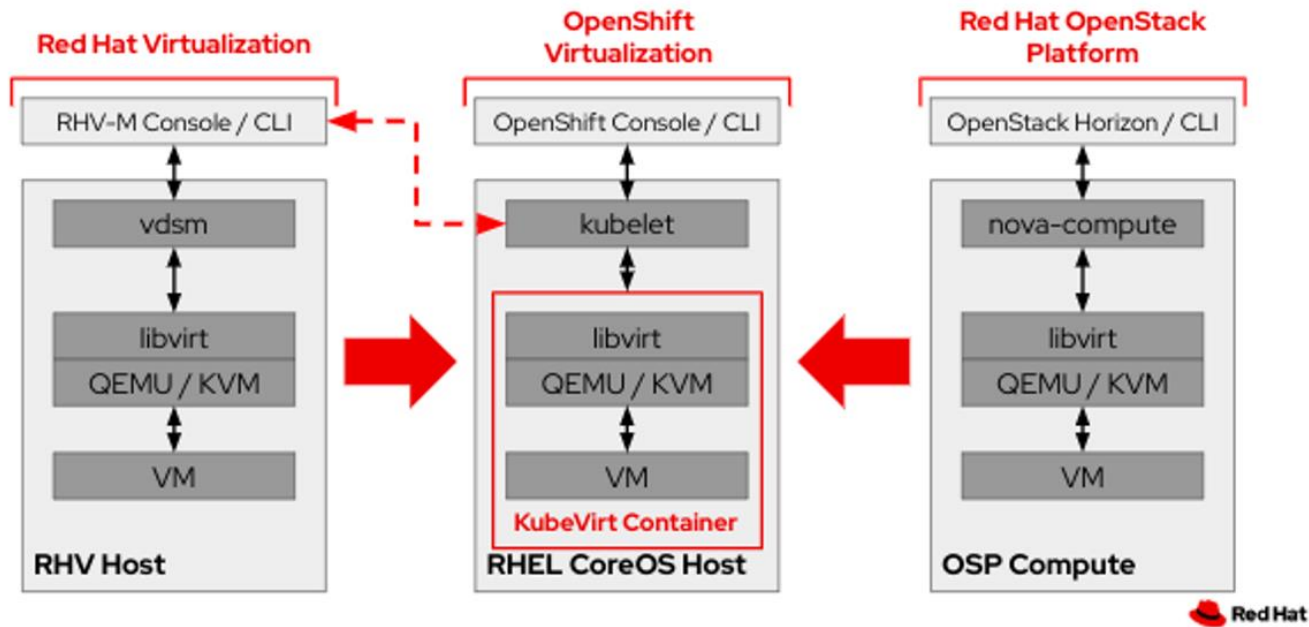
OpenShift Virtualization

Red Hat OpenShift Container Platform

Red Hat Enterprise Linux CoreOS

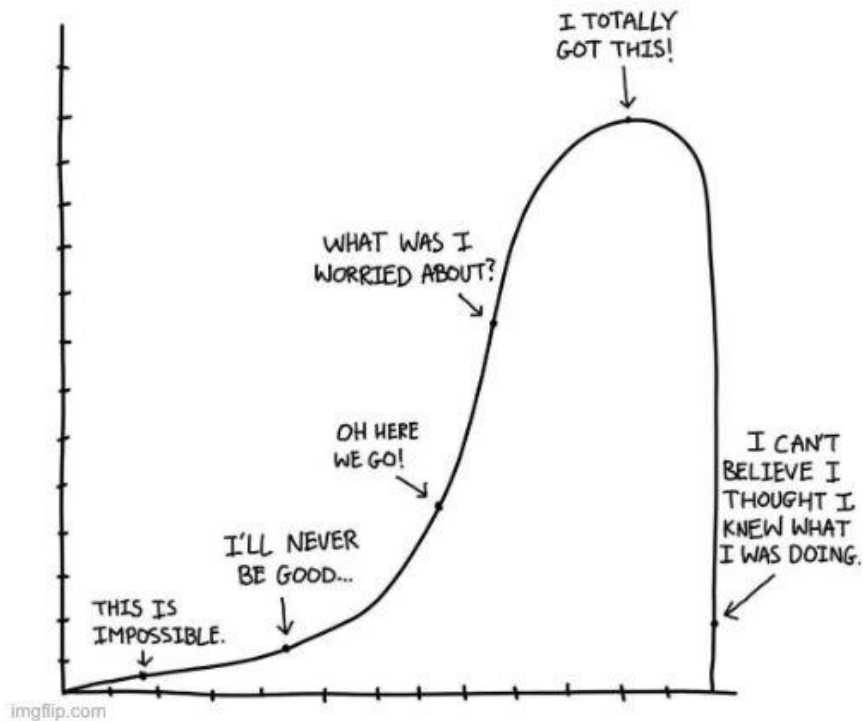
Physical machine

OpenShift Virtualization is based on the “Container-Native-Virtualization” technology

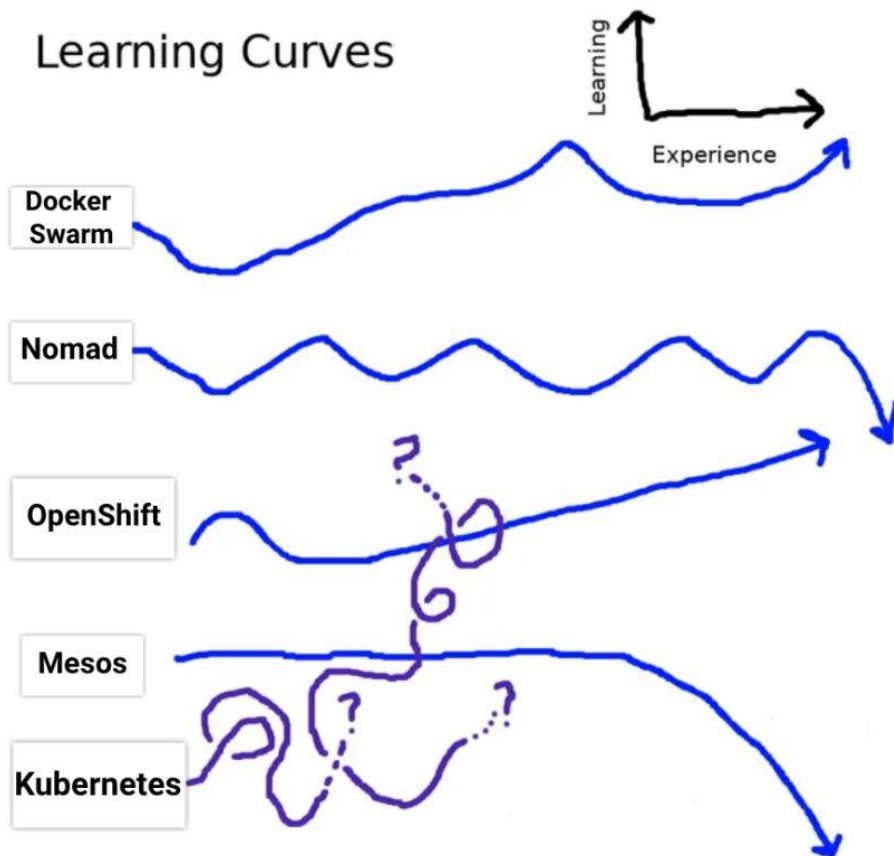


THE KUBERNETES LEARNING CURVE

LEARNING CURVE



Learning Curves



Solved all your problems. You're welcome.

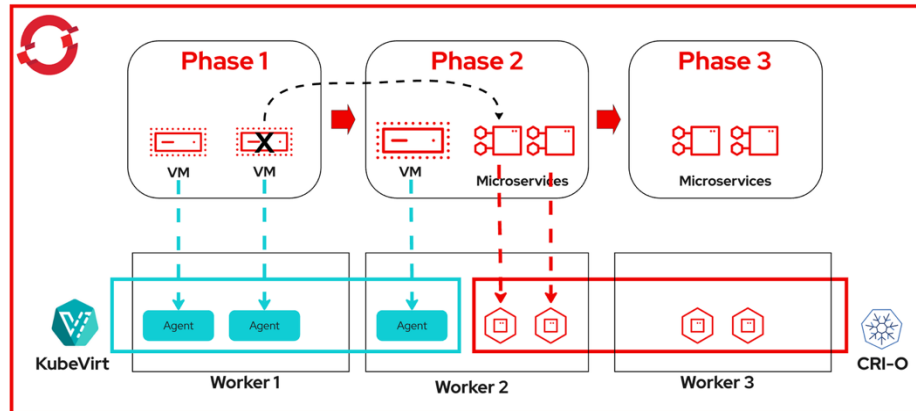




WHY OPENSIFT VIRTUALIZATION?

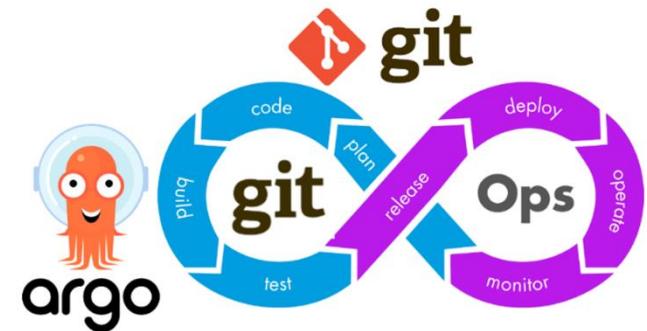
APP MODERNISIERUNG

- VM Workload parallel zur Container Workload
- Migration klassischer Workload zu Container Workload
- Nutzen von Kubernetes Features für die VM Welt
 - K8S Service Loadbalancer
 - Health Checks
 - Secret / ConfigMaps mounts



BUILD IN AUTOMATISIERUNG MIT GITOPS

- Config Lifecycle der Plattform.
Bsp.: Netzwerk Config, VM Templates, Image Imports
- Komplexe Automatisierungen.
Bsp.: pull-request -> Aufbau der kompletten Test Infrastruktur
- Revision sicher & YAML



APP HEALTH

Progressing

SYNC STATUS

OutOfSync from HEAD (0da8bf8)

Auto sync is not enabled.

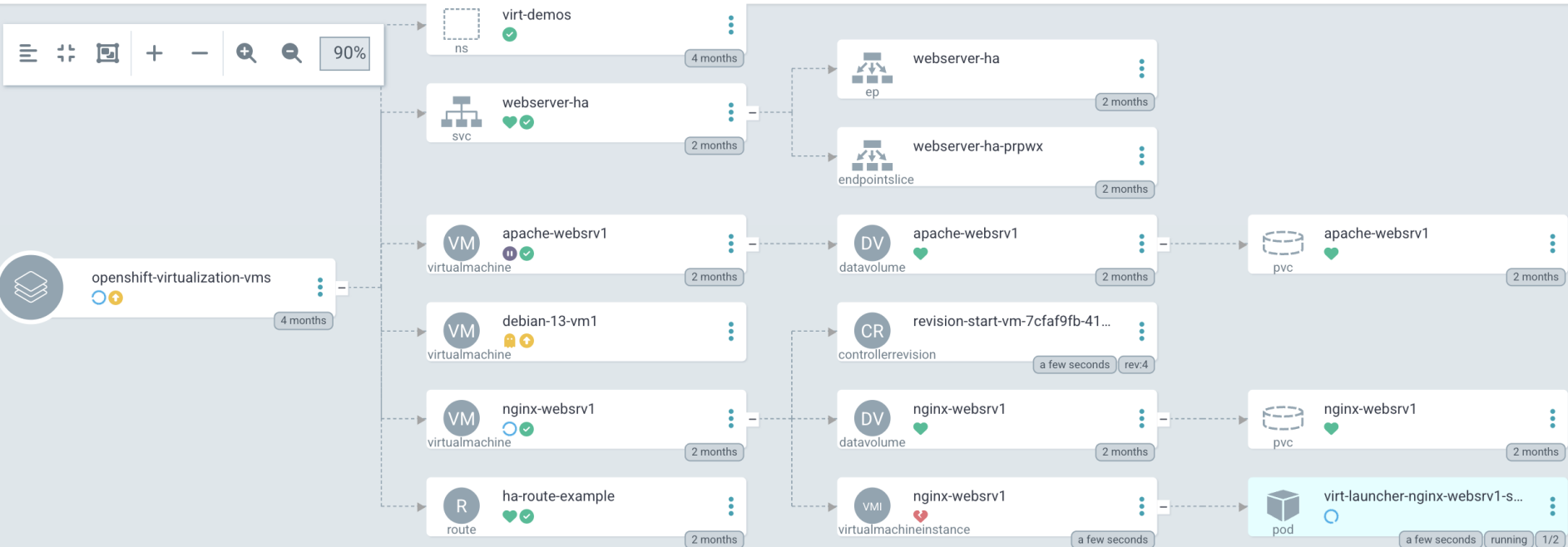
Author: Andreas Tomaselli <andreas.tomaselli@nts.eu> -
Comment: feat: added trident https://access.redhat.com/solutio...

LAST SYNC

Sync OK to b4eb6f5

Succeeded 2 months ago (Tue Jan 13 2026 08:44:37 GMT+0100)

Author: Andreas Tomaselli <andreas.tomaselli@nts.eu> -
Comment: fix: replaced cloud-init stuff



GEMEINSAME PLATFORM

- Openshift integriert sich tiefer in bestehende Netzwerk / Storage Lösungen
- Unified Teams, gleicher Software Toolstack
- Moderne Plattform für moderne Probleme
 - Compliance / Observability
 - AI/ML
 - Automatisierung
 - Security
 - OCI Containers (VMs)

GEMEINSAME PLATFORM

```
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataSource
metadata:
  name: debian-13
  namespace: openshift-virtualization-os-images
spec:
  source:
    pvc:
      name: debian-13
      namespace: openshift-virtualization-os-images
status:
  conditions:
    - lastHeartbeatTime: '2026-02-25T15:53:46Z'
      lastTransitionTime: '2026-02-25T15:53:46Z'
      message: DataSource is ready to be consumed
      reason: Ready
      status: 'True'
      type: Ready
  source:
    pvc:
      name: debian-13
      namespace: openshift-virtualization-os-images
```

```
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataVolume
metadata:
  name: debian-13
  namespace: openshift-virtualization-os-images
spec:
  source:
    http:
      url: 'http://172.28.88.139/external/debian-13-generic-amd64.qcow2'
  storage:
    resources:
      requests:
        storage: 30Gi
status:
  claimName: debian-13
  conditions:
    - lastHeartbeatTime: '2025-11-23T09:24:12Z'
      lastTransitionTime: '2025-11-23T09:24:12Z'
      message: Import Complete
      reason: Completed
      status: 'False'
      type: Running
  phase: Succeeded
  progress: 100.0%
```



Red Hat
OpenShift
Platform Plus



Red Hat
OpenShift
Container Platform



Red Hat
OpenShift
Kubernetes Engine



Red Hat
OpenShift
Virtualization Engine



OpenShift Virtualization

Provided by Red Hat



Install

Channel

stable

Version

4.20.7

Capability level

- ☒ Basic Install
- ☒ Seamless Upgrades
- ☒ Full Lifecycle
- ☒ Deep Insights
- ☐ Auto Pilot

Source

Red Hat

Provider

Red Hat

Requirements

Your cluster must be installed on bare metal infrastructure with Red Hat Enterprise Linux CoreOS workers.

Details

OpenShift Virtualization extends Red Hat OpenShift Container Platform, allowing you to host and manage virtualized workloads on the same platform as container-based workloads. From the OpenShift Container Platform web console, you can import a VMware virtual machine from vSphere, create new or clone existing VMs, perform live migrations between nodes, and more. You can use OpenShift Virtualization to manage both Linux and Windows VMs.

The technology behind OpenShift Virtualization is developed in the [KubeVirt](#) open source community. The KubeVirt project extends [Kubernetes](#) by adding additional virtualization resource types through [Custom Resource Definitions](#) (CRDs). Administrators can use Custom Resource Definitions to manage [VirtualMachine](#) resources alongside all other resources that Kubernetes provides.

MIGRATION TOOLKIT FOR VIRTUALIZATION (MTV)

Plan Migration
Plan Migration

Migration plans ③

Filter	Name	Filter by name	→	Show archived	Manage columns
Name ↑	Source provider	Virtual machi...	Migration status	Migration type	Migration star...
PL mig-plan	vm fdh-vmware	1 VM	▶ Start	Warm	-

Run Migration

Migration plans ③

Filter	Name	Filter by name	→	Show archived	Manage columns
Name ↑	Source provid...	Virtual ...	Migration status	Migration type	Migration started
PL mig-plan	vm fdh-vmware	1 VM	16% ⚠ 1	Warm	Schedule cutover 🌐 16. März 2026, 21:12

MIGRATION TOOLKIT FOR VIRTUALIZATION (MTV)

Run Migration

	Name ↑	Pipeline status ↑	Disk transfer	Disk counter	Started at
▼ ☐	atom-debian-vm1	⌛ Running	9338 / 16384 MB	0 / 1 Disks	16. März 2026,

▼ Migration progress Warm

Name	Description	Completed at
✓ Initialize	Initialize migration.	16. März 2026, 21:12
✓ PreflightInspection	Inspect VM before migration.	16. März 2026, 21:14
○ DiskTransfer	Completed 0 of 1 DiskTransfer tasks	-
○ Cutover	Paused Schedule cutover	-
○ ImageConversion	Convert image to kubvirt.	-
○ VirtualMachineCreation	Create VM.	-



OPENSIFT VIRTUALIZATION NETWORKING

OPENSIFT – OVN-KUBERNETES CNI

- **NMState Operator**
- **POD Networking**
- **CUDN**
- **Localnet**
- **BGP Routing (4.19.12+ nativ)**

NMSTATE – DEKLARATIVE NETWORK CONFIG

```

apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: virt-node1.bond1-ex-network
spec:
  desiredState:
    interfaces:
      - ipv4:
          dhcp: false
          enabled: false
        link-aggregation:
          mode: 802.3ad
          options:
            lacp_rate: fast
            xmit_hash_policy: layer2+3
          port:
            - enp10s0
            - enp11s0
          mtu: 1500
          name: bond1
          state: up
          type: bond
    nodeSelector:
      kubernetes.io/hostname: atom-os-cl1-node1

```

```

apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: ovs-br1-multiple-networks
spec:
  desiredState:
    interfaces:
      - bridge:
          allow-extra-patch-ports: true
          options:
            mcast-snooping-enable: true
            stp: false
          port:
            - name: bond1
          name: ovs-br1
          state: up
          type: ovs-bridge
    ovn:
      bridge-mappings:
        - bridge: ovs-br1
          localnet: localnet-fdh
          state: present
    nodeSelector:
      virtualization.host: ''

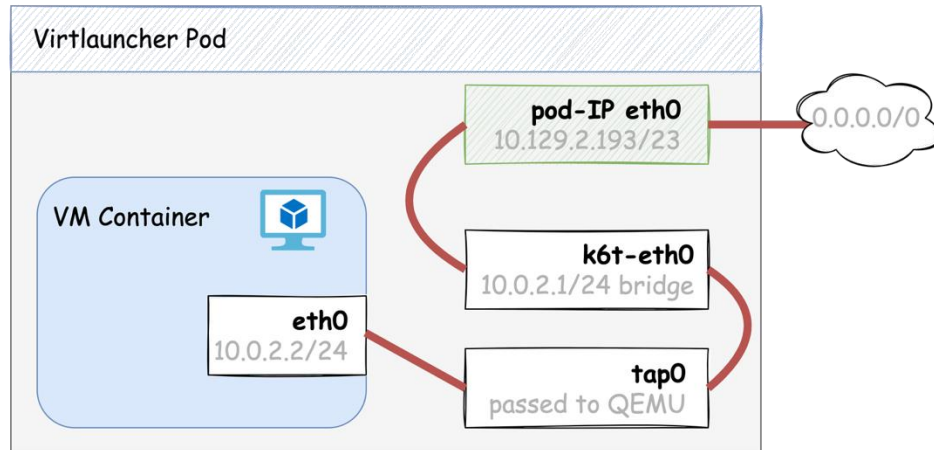
```

NMSTATE – DEKLARATIVE NETWORK CONFIG

```
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: infra1-if-ens224-bgp
spec:
  desiredState:
    interfaces:
      - ipv4:
          address:
            - ip: 192.168.123.101
              prefix-length: 24
          enabled: true
          name: ens224
          state: up
          type: ethernet
    nodeSelector:
      kubernetes.io/hostname: atom-os-cl1-infra1
```

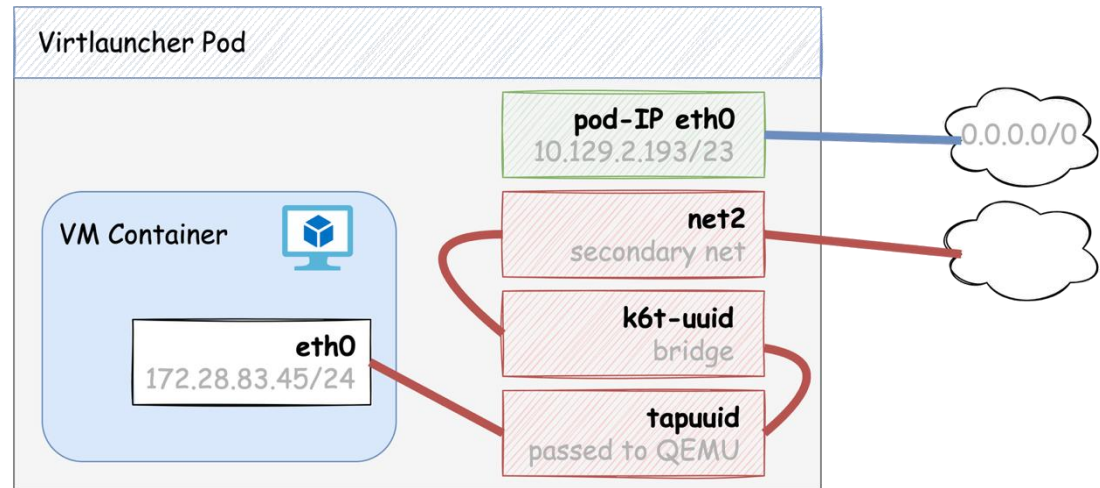
POD NETWORK

- Bridge innerhalb des Pods liefert TAP Device an die VM
- VM IP **masquerade** via Pod-IP
- Pod IP ändert sich mit VM-Migration
- VM IP ist dauerhaft 10.0.2.2/24 !!



POD NETWORK / VM SECONDARY / LOCALNET

- Secondary Netzwerk kann via Multus und NAD angehängt werden.
- No IP Pod Net2 , pass-through an die VM
- eth0 default route



MICROSEGMENTIERUNG

- **NetworkPolicy**
nur für das Primary Pod Network

MICROSEGMENTIERUNG

- **NetworkPolicy**
nur für das Primary Network
- **MultiNetworkPolicy**
für Secondary Pod Networks
 - Localnet nur IP & Ports
 - Kein Label support

```
apiVersion: k8s.cni.cncf.io/v1beta1
kind: MultiNetworkPolicy
metadata:
  name: demo-sample-ingress
  namespace: virt-demos
  annotations:
    k8s.v1.cni.cncf.io/ policy-for: virt-demos/vlan-83
spec:
  podSelector: {}
  policyTypes:
    - Ingress
  ingress:
    - from:
        - ipBlock:
            cidr: 172.28.80.0/24
            except: []
        - ipBlock:
            cidr: 172.28.81.0/24
            except: []
      ports:
        - port: 80
          protocol: TCP
      # from mgmt network to ssh service
    - from:
        - ipBlock:
            cidr: 172.28.88.0/24
            except: []
      ports:
        - port: 22
          protocol: TCP
```

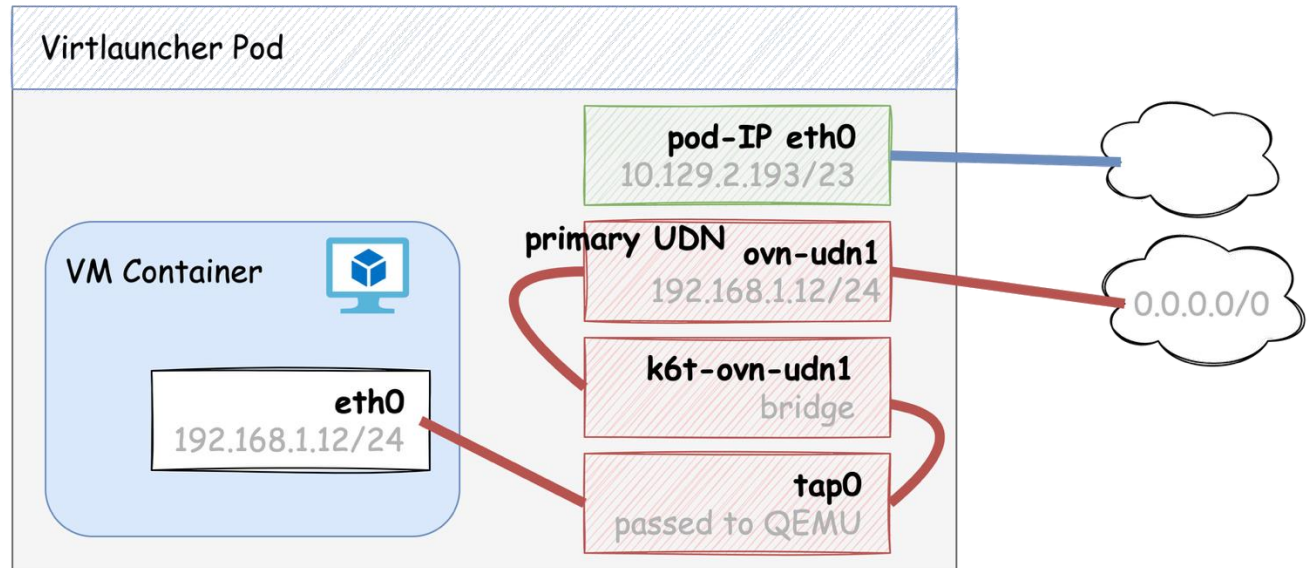

USER DEFINED NETWORKS (UDN & CUDN)

- UDN
 - Namespace scoped
 - User created (non C-Admin)
 - Kein VLAN Support
- CUDN
 - Cluster scoped
 - Erstellt von Cluster Admins
 - VLAN Support
 - Shared Network mehrere Namespaces
 - BGP Advertising möglich

UDN TOPOLOGY

- Layer2
 - Overlay logical switched zwischen Nodes
 - Supported VMs
 - Primary & Secondary Network
- Layer3
 - Overlay routed zwischen Nodes
 - Kein VM Support (no Live Migration)
 - Primary & Secondary Network
- Localnet
 - VLAN durchreichen für klassische Netzwerke
 - VM support
 - Secondary Network limitiert
 - CUDN Only

POD NETWORK – PRIMARY UDN



CUDN - LOCALNET

```

apiVersion: k8s.ovn.org/v1
kind: ClusterUserDefinedNetwork
metadata:
  name: fdh-vlan83
spec:
  namespaceSelector:
    matchLabels:
      cluster-udn: vlan83
  network:
    localnet:
      ipam:
        mode: Disabled
      physicalNetworkName: localnet-fdh
      role: Secondary
      vlan:
        access:
          id: 83
          mode: Access
        topology: Localnet
status:
  conditions:
    - lastTransitionTime: '2025-11-22T19:09:22Z'
      message: 'NetworkAttachmentDefinition has been created in following namespaces: [virt-demos, virt-demos2]'
      reason: NetworkAttachmentDefinitionCreated
      status: 'True'
      type: NetworkCreated

```

```

apiVersion: v1
kind: Namespace
metadata:
  creationTimestamp: "2025-11-22T17:23:33Z"
  labels:
    cluster-udn: vlan83
    kubernetes.io/metadata.name: virt-demos
  name: virt-demos
spec:
  finalizers:
    - kubernetes
status:
  phase: Active

```

CUDN – PRIMARY L2

```
apiVersion: k8s.ovn.org/v1
kind: ClusterUserDefinedNetwork
metadata:
  name: primary-udn-l2
spec:
  namespaceSelector:
    ✓ matchLabels:
      udn: l2-primary
  network:
    layer2:
      ipam:
        lifecycle: Persistent
        role: Primary
        subnets:
          - 10.110.0.0/24
    topology: Layer2
```

BGP SUPPORT

- Seit Openshift 4.19.12 nativ via ffr-k8s Project im Cluster Network Operator integriert via **RouteAdvertisements CR**
- Advertise CUDN Prefixes (L2 & L3 , VRF)
- Advertise EgressIPs
- Advertise Pod Network
- Whats Next
 - BGP EVPN VXLAN Support (bring your own fabric)

CISCO – ACI CNI / CISCO NETWORK OPERATOR

- Native Implementierung in Cisco ACI via Opflex
 - Pods 1:1 Mapping vorhandene EPG/ESG
 - Enforcement via ACI Fabric
 - Ersetzt das Primary CNI
- Cisco Network Operator
 - Erstellen von ACI Objecten (EPG/BD/Peerings) via CR
 - Primary CNI wird nicht ersetzt
 - Chainload möglich
 - **VMM Lite Feature mit CNI 6.1.5**
 - Erstellen von NAD innerhalb von Openshift Virtualization

```

apiVersion: aci.fabricattachment/v1
kind: NetworkFabricConfiguration
metadata:
  name: networkfabricconfiguration
  namespace: aci-containers-system
spec:
  vlans:
    - epg:
        tenant: LAB-GRZ-DEFAULT
        applicationProfile: DEMO-APP
        bd:
          common-tenant: false
          name: APP1-BD
          subnets:
            - subnet: '192.168.11.1/24'
          vrf:
            common-tenant: false
            name: LAB-GRZ-DEFAULT
        contracts:
          consumer:
            - allow-any
            - my-special-contract
          provider:
            - http-any
        lldpDiscovery: true
        name: APP1-EPG
      vlans: "101"

```


CNO- VMM LIGHT FOR OPENSIFT VIRTUALIZATION

FDH-LAB

- Quick Start
- FDH-LAB
 - Application Profiles
 - OCP-VIRT
 - Application EPGs
 - VLAN0083**
 - Domains (VMs and Bare-Metals)
 - EPG Members
 - Static Ports
 - Static Leafs
 - Fibre Channel (Paths)
 - Contracts
 - Static Endpoint

EPG - VLAN0083

Summary **Policy** Operational Stats

Topology **General**

Properties

Name: VLAN0083

Alias:

Description: optional

Annotations: cno-namespaces ☒

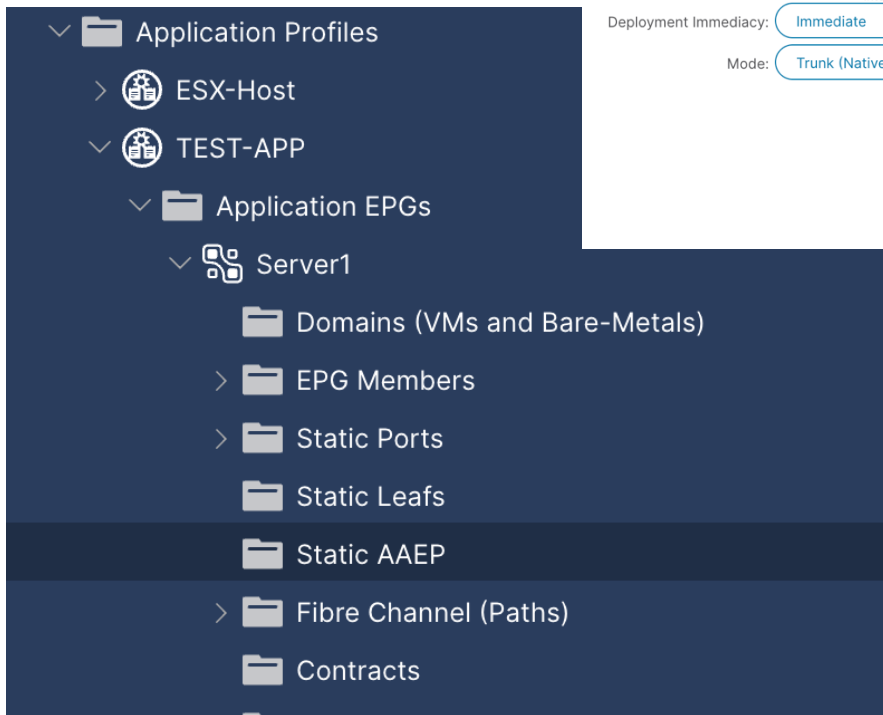
Global Alias:

Application EPGs

Application EPGs	Encap	Primary Encap	Mode
FDH-LAB/OCP-VIRT/VLAN0083	vlan-3600	unknown	Trunk

AAEP Policy

ACI 6.1.5+



Create Static AAEP

Name: !

Encap:

Primary Encap:

Deployment Immediacy: ☒ ☐

Mode: ☐ ☒ ☐

NAD fdh-lab-ocp-virt-vlan0083-d927bbf5ba367782

Details YAML

```

1  apiVersion: k8s.cni.cncf.io/v1
2  kind: NetworkAttachmentDefinition
3  metadata:
4    annotations:
5      aaep-name: AEP_VMM
6      aci-sync-status: in-sync
7      cno-name: ''
8      epg-dn: uni/tn-FDH-LAB/ap-OCP-VIRT/epg-VLAN0083
9      managed-by: cisco-network-operator
10   vlan: '3600'
11   resourceVersion: '77174692'
12   name: fdh-lab-ocp-virt-vlan0083-d927bbf5ba367782
13   uid: 4086e2a4-4952-4f04-a12d-905de5924bf9
14   creationTimestamp: '2026-01-12T10:27:03Z'
15   generation: 1
16 > managedFields: ""
39   namespace: virt-demos
40   labels:

```

INTEGRATION

NTS

VirtualMachines

VirtualMachine details

VM atom-debian-vm1 Running

YAML

Actions

Overview

Metrics

YAML

Configuration

Events

Console

Snapshots

Diagnostics

Details

Storage

Network

Scheduling

SSH

Initial run

Metadata

Network interfaces

Add network interface

Filter

Name

Search by name...

Name	Model	Network	State	Type	MAC add...
net-0	virtio	fdh-lab-ocp-virt-vlan0083-d927bbf5ba367782		Bridge	00:50:56:93:4c:d3

FDH-LAB

Quick Start

FDH-LAB

Application Profiles

OCP-VIRT

Application EPGs

VLAN0083

Domains (VMs and Bare-Metals)

EPG Members

Static Ports

Static Leafs

EPG - VLAN0083

Summary

Policy

Client Endpoints

Configured Access Policies

MAC/IP	Endpoint Name	Learning Source
00:50:56:93:4C:D3	172.28.83.45	Learned

Informationsklasse: Public/Internal/Confidential/Restricted (Unzutreffendes löschen: Einfügen > Kopf- und Fußzeile > ... > Für alle übernehmen)